Public-Data Hedging Benchmark

Approach, current results, and model catalogue

Status date: 13 October 2026

Executive status. The model-neutral daily benchmark is operational on a normalized, original-data-free public-candidate panel. Two methods have complete out-of-sample results over all 611 eligible 2023–2025 origins and all 20 daily digital targets. SANOS–LV reduces aggregate pure-hedge RMSE by **56.5%** versus the unhedged claims; historical statistical SANOS reduces it by **17.4%**. Robust-covariance production is in progress. The first rolling GROMM market fit is complete under Q, but its compatible-P drift is rejected; bounded normalization of float32 decode roundoff now makes the authenticated Q artifact operational. VolGAN and AD–Seq–Vol pass their current implementation gates but do not yet have comparable full-production results. Weekly mechanics and pricing pass. The 94-case weekly matrix is materialized: 28 native CPU cases are ready and deferred until the active daily queues finish, four generated-model cases await their review gate, and 62 cases remain blocked on incomplete authenticated artifacts.

1 Approach

1.1 Data boundary and common market

The sole runtime input is the distributable-candidate dataset `spx-2022-2025-hedging-public-candidate-v1`. It contains 891 retained dates from 3 January 2022 to 29 August 2025 and 2,014,740 contemporaneously eligible quote-side option coordinates. It contains no original WRDS price, midpoint, identifier, volume, open interest, vendor Greek, or vendor implied volatility. Instead it releases normalized fitted SANOS and DLV states, separate reconstructable SANOS and DLV model prices, Black implied volatilities, forward/spot ratios, discount factors, normalized half-spreads, and forward-adjusted spot returns.

The common executable market mark is the released SANOS price. A purchase pays SANOS plus one half-spread and a sale receives SANOS minus one half-spread. Spot costs one basis point of traded notional per transaction; cash settlement is free. The release is technically classified as `public_candidate`; redistribution still requires a separate rights decision.

1.2 Out-of-sample protocol

- Model initialization, architecture validation, and target sizing use 2022 only. At each date t, every rolling estimator uses the current market state and the trailing 252 exact one-session transitions available at t, with no future observation.
- Daily evaluation runs from 17 January 2023 through 28 August 2025: 611 eligible origins, 20 targets per origin, and 12,220 claim/origin records.
- The targets are cash digitals at 1D and 3D: ATM call and put plus OTM NMS levels $\pm 0.25, \pm 0.50, \pm 0.75, \pm 1.00$. Quantities are fixed from 2022 so each target has one maturity-matched ATM spot-risk unit.
- The hedge universe is normalized spot plus every eligible released market strike strictly inside $(-2, 2)$ contemporaneous NMS, represented by its OTM option (put below the forward and call at or above it). It uses every expiry up to five business days, or the first later expiry when a five-day expiry is absent.
- Weekly evaluation starts an overlapping five-session episode on every eligible date. Frozen weekly fits once and keeps its inception contract book; restarted weekly refits daily. Both may react to the realized state.

1.3 Common accounting and objectives

Every method starts from the same market claim mark and is evaluated on the same realized next-date SANOS mark or cash payoff. The report separates *pricing loss*, *pure hedge loss*, and their sum, *economic loss*. Buyer and seller indifference prices are additional model outputs, not substitutes for this common-price comparison.

Supported risk criteria are mean–variance, mean–volatility, entropy OCE, and power-exponential OCE, following the risk-functional hedging framework of Buehler et al. [ ]. Learned or statistical policies use the hedge of the claim relative to the corresponding zero-claim policy; generated models also fit a direct pure policy so transaction costs are charged on the executable net position rather than inferred from differences of absolute-policy costs.

2 Current Results

2.1 Completed daily production runs

The four objective tables evaluate each completed policy on its full 611-origin out-of-sample *economic loss*, including transaction costs and the common market-price contribution. Risk functionals are evaluated before rescaling; only the final scalar is displayed in basis points of inception normalized spot. Each entry is the equal-weighted mean of the 20 target-level scores. The historical row is the single fitted mean–volatility $\kappa = 1$ policy evaluated under all four functionals; it is not presented as four separately refitted policies. As the remaining authenticated production runs finish, they enter as additional fitted-policy rows.

Table 1: Mean–variance: full out-of-sample economic-risk score.

Model	Fitted policy	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	Book MDD	Worst target MDD
SANOS–LV	reference	11.18	11.39	13.58	6855.28	15649.50
Historical SANOS	MVol, $\kappa = 1$	7.66	8.48	16.61	5290.14	8024.31

Table 2: Mean–volatility: full out-of-sample economic-risk score.

Model	Fitted policy	$\kappa = 0.1$	$\kappa = 1$	$\kappa = 2$	$\kappa = 3$	Book MDD	Worst target MDD
SANOS–LV	reference	17.83	77.96	144.76	211.56	6855.28	15649.50
Historical SANOS	MVol, $\kappa = 1$	20.16	133.44	259.30	385.17	5290.14	8024.31

Table 3: Entropy OCE: full out-of-sample economic-risk score.

Model	Fitted policy	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	Book MDD	Worst target MDD
SANOS–LV	reference	11.18	11.39	13.57	6855.28	15649.50
Historical SANOS	MVol, $\kappa = 1$	7.66	8.56	48.84	5290.14	8024.31

Table 4: Power-exponential OCE: full out-of-sample economic-risk score.

Model	Fitted policy	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	Book MDD	Worst target MDD
SANOS–LV	reference	11.18	11.39	13.52	6855.28	15649.50
Historical SANOS	MVol, $\kappa = 1$	7.66	8.47	16.53	5290.14	8024.31

Table 5 gives complementary distributional diagnostics. Aggregate RMSE is the square root of the equally weighted mean claim-level MSE; every claim contributes 611 observations. "Reduction" is relative to the matching unhedged RMSE. Expected shortfall is the equal-weighted target-level 95% expected shortfall. Book MDD is the maximum drawdown of the cumulative P&L of an equal-weighted 20-target daily book; worst target MDD is the largest single-target drawdown. Drawdowns start from zero and are reported in basis points of inception normalized spot, not as percentages of an unspecified capital account.

Table 5: Out-of-sample daily distributional risk, in basis points except reduction.

Model	Targets	Unhedged RMSE	Pure RMSE	Reduction	ES_{95}	Book MDD	Worst MDD
SANOS–LV	all 20	163.16	70.96	56.5%	135.07	6855.28	15649.50
SANOS–LV	1D	124.40	86.20	30.7%	132.86	3398.88	5483.11
SANOS–LV	3D	194.33	51.37	73.6%	137.27	10641.26	15649.50
Historical SANOS	all 20	163.16	134.69	17.4%	257.10	5290.14	8024.31
Historical SANOS	1D	124.40	111.59	10.3%	223.36	5247.75	8024.31
Historical SANOS	3D	194.33	154.36	20.6%	290.85	5338.99	7356.66

The SANOS–LV hedge is materially stronger on this first completed comparison, especially for the remaining-life 3D claims. Its aggregate pure RMSE is 47.3% lower than historical SANOS. Pricing loss is negligible at aggregate scale: SANOS–LV economic RMSE is 70.98 bp versus pure RMSE 70.96 bp; historical SANOS uses the common price and therefore has zero pricing contribution by design.

Table 6: Average trading diagnostics across all 20 targets and 611 origins.

Model	Mean cost (bp)	Gross marked exposure	Cap hits	Nonconverged	Records
SANOS–LV	10.11	267.0%	0	0	12,220
Historical SANOS	10.19	7.4%	0	0	12,220

Raw support counts are deliberately omitted. Conic-solver residuals and the difference between claim and zero-claim policies make a numerical "nonzero" threshold economically arbitrary. Table 6 instead reports the actual paid cost and the origin-marked gross executable position. SANOS–LV's larger marked exposure reflects its spot delta and exact factor hedge; the historical policy pays a similar average cost on a much smaller marked book.

2.2 Evidence level across models

Table 7: Current evidence. "Readiness" is not a performance estimate.

Model	Measure	Daily status	Evidence
SANOS–LV	Q price / realized path	**complete**	611 origins, 12,220 records, zero cap hits and zero nonconvergence.
Historical SANOS	P	**complete**	611 origins, 12,220 records, zero cap hits and zero nonconvergence.
Robust covariance	P	**pending**	Full 20-target one-origin mean–volatility gate passed for $\kappa \in \{0.1, 1, 2, 3\}$; the 611-origin production run is in progress.
GROMM	Q only	**ready**	First rolling Q calibration completed; compatible P failed its fixed gates, while authenticated real-artifact Q replay produces finite scenarios.
VolGAN	P	**ready**	Public-architecture implementation, restartable rolling updates, daily/weekly adapters, and smoke tests pass; production pending.
AD–Seq–Vol / FT	P	**ready**	Full-geometry managed-GPU origin: 4,096 scenarios, 20 targets, 158.3 seconds, 2,636 MiB VRAM, no cap hit or nonconvergence; production pending.

2.3 Supporting calibration and weekly evidence

At the first GROMM-eligible origin, 17 January 2023, the 200-strike affine market fit places 84.90% of observations inside their released spread. The rank-16 DLV innovation basis explains 95.15% of fitted innovation variance. The compatible-P projection explains only 0.633% of its target energy and its holdout loss improves by 0.473%, below the fixed 50% and 0.5% gates. The gain statistic is 9.54 and the daily risk-price norm is 0.102, but those two diagnostics do not override the failed acceptance gates. The artifact is therefore Q-only. Its float32 serialized densities miss unit mass by at most 1.964×10^{-8}; the loader now validates that residual against the stored precision, normalizes in float64 to 2.22×10^{-16} maximum error, and rejects larger deviations. An authenticated four-path replay produced finite Q returns over 163 instruments and rejected P as required. These are calibration and implementation diagnostics, not hedge-performance results.

The weekly engine has 528 eligible origins through 22 August 2025 and 13 common targets: ten five-day digitals, realized variance, a 95% end-of-day down-and-out ATM put, and a 105% end-of-day up-and-out ATM call. The common DLV transition test has maximum mass and martingale errors below 3.6×10^{-14} across an eight-date check. Daily, frozen-weekly, and restarted-weekly adapters exist for all model families, but no complete weekly comparison is yet reported. The production matrix contains 94 cases. Its 28 ready SANOS–LV and historical-SANOS CPU cases are queued behind the active daily CPU runs; four generated-model cases await their production gate, and 62 GROMM or generated-model cases remain explicitly blocked rather than being launched without complete evidence.

3 Model Catalogue

3.1 SANOS–LV

SANOS supplies the common market interpolation and digital price [ ]. DLV local volatility supplies the model price and Greeks. From each trailing 252-day window, the method estimates one front-spot-projected DLV factor and seven residual PCA factors. Central DLV-squared bumps give factor vegas; a linear program selects the minimum predictable-cost option portfolio that exactly neutralizes them, and spot removes residual delta. One-day claims are delta-only; three-day claims use at most eight option hedges. The method is deterministic, sparse, and the strongest

completed baseline.

3.2 Historical SANOS

Historical SANOS reconstructs 252 same-coordinate option and target returns from released SANOS states [2] and uses those empirical rows directly as scenarios. It supports all four risk criteria in the deep-hedging sense of Buehler et al. [1]. Long, short, and zero policies determine indifference prices; a separate drift-neutral policy gives the executable pure hedge. Mean–variance uses exact cost-lasso coordinate descent, mean–volatility uses an exact conic program with an explicit recession test, and entropy and power-exponential objectives use analytic-gradient L-BFGS-B. The model is inexpensive and fully rolling, but its completed mean–volatility $\kappa = 1$ result trails SANOS–LV.

3.3 Robust covariance

This baseline deliberately avoids a SANOS scenario generator. It replays the current relative option book over 252 historical public transitions, fits robust scales and a regularized Tyler scatter matrix [3, 5], and estimates each target cross-moment separately by robust regression. Expected returns are HAC-screened in covariance eigenportfolios. Mean–variance and mean–volatility hedges use spread-weighted lasso costs; the latter is checked analytically for recession directions. The implementation gate passes and a full production run is in progress, but incomplete outputs must not be compared with Table 5.

3.4 GROMM

GROMM is the native generative risk-neutral option-market model: a 200-strike JAX market fit, 16-factor AR(1) log-DLV-squared dynamics, independent 5% Black background, and eight Brownian substeps per business day. Dated artifacts are bound to exactly 252 public transitions and must pass compatible-P drift gates. The first rolling fit clears the market-fit threshold but rejects the compatible P measure, so it supplies Q scenarios only. Bounded float32 decode normalization and authenticated Q replay now pass. An affine-plus-deep LSTM supports all four objectives; whenever P is accepted, the executable pure policy uses conditional Q-shadow gains at each realized P state, removing predictable physical drift without fitted holdout centering. Full rolling production remains outstanding. A separate conventional Q comparator prices by risk-neutral expectation and computes spot delta plus eight contemporaneous DLV-factor vegas from the same stochastic GROMM state using fixed-seed common random numbers [1].

3.5 VolGAN

VolGAN is a historical-P surface generator on a fixed 9×11 Black-IV grid reconstructed from public SANOS states. It conditions on two lagged returns, trailing 21-day realized volatility, and the current surface. The published two-layer Softplus architecture uses 32 noise dimensions, a 16-unit hidden layer, and strike/maturity gradient penalties. Weekly paths recursively update surface and return state. The implementation is project-native and uses no proprietary surfaces. Smoke, restart, and rolling warm-start gates pass; full production does not yet exist [6].

3.6 AD–Seq–Vol / FT

AD–Seq–Vol is a causal sequence diffusion model on 22-day histories of the same 9×11 surface plus forward-adjusted returns. Its eight-layer, width-256, four-head transformer predicts the clean sequence through a 200-step cosine DDPM. AD–Seq–Vol–FT freezes the EMA base and trains a zero-initialized rank-four adapter with calendar, call-spread, and butterfly penalties. Both are historical-P generators and supply indifference prices rather than Q expectations. Exact restart, rolling warm-start, daily/weekly adapters, and one full-geometry GPU origin pass; production and batch-invariance remain pending. The architecture and published comparison are due to Han et al. [4].

Provenance. The completed results use public dataset manifest SHA-256 `58741552e0f6...`.
SANOS–LV run identity begins `2af7a4f64df4`; historical-SANOS run identity begins `64adf168a213`.
All headline numbers are derived from immutable `hedging-daily-results-v2` artifacts.

A Target-Level Out-of-Sample Objectives

Each table reports the complete 611-origin score for every target and completed policy. Objectives are evaluated on the original inception-spot-normalized economic loss, including transaction costs, and the final scalar alone is multiplied by 10^4 for display in basis points. Pure RMSE and economic maximum drawdown use the same target notional. This preserves the risk-aversion parameters shown in the column headings; applying them after converting losses to basis points would be dimensionally incorrect.

A.1 Mean–variance

Table 8: Target-level mean–variance scores and drawdowns.

Model/policy	Target	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	Pure RMSE	Economic MDD
SANOS–LV (reference)	1D put −1.00 NMS	5.63	5.93	8.97	82.35	4172.04
SANOS–LV (reference)	1D put −0.75 NMS	6.41	6.72	9.79	82.88	4684.10
SANOS–LV (reference)	1D put −0.50 NMS	3.21	3.49	6.31	79.28	3716.53
SANOS–LV (reference)	1D put −0.25 NMS	4.41	4.68	7.37	77.61	3536.83
SANOS–LV (reference)	1D put ATM	4.04	4.33	7.29	81.19	4226.89
SANOS–LV (reference)	1D call ATM	1.75	2.04	5.00	81.02	1877.49
SANOS–LV (reference)	1D call +0.25 NMS	4.95	5.29	8.71	87.25	3459.89
SANOS–LV (reference)	1D call +0.50 NMS	6.73	7.11	10.96	92.56	4482.35
SANOS–LV (reference)	1D call +0.75 NMS	7.90	8.31	12.46	96.20	5483.11
SANOS–LV (reference)	1D call +1.00 NMS	4.23	4.67	9.05	98.79	4163.14
SANOS–LV (reference)	3D put −1.00 NMS	12.15	12.25	13.19	47.57	7418.17
SANOS–LV (reference)	3D put −0.75 NMS	11.17	11.25	12.05	43.92	6817.57
SANOS–LV (reference)	3D put −0.50 NMS	20.03	20.13	21.10	50.80	12232.64
SANOS–LV (reference)	3D put −0.25 NMS	25.62	25.73	26.78	54.46	15649.50
SANOS–LV (reference)	3D put ATM	22.73	22.83	23.83	52.08	14016.55
SANOS–LV (reference)	3D call ATM	21.98	22.08	23.05	51.50	13426.36
SANOS–LV (reference)	3D call +0.25 NMS	22.23	22.33	23.32	51.87	13590.55
SANOS–LV (reference)	3D call +0.50 NMS	13.98	14.07	14.97	46.72	8542.56
SANOS–LV (reference)	3D call +0.75 NMS	10.83	10.94	12.01	49.78	6866.03
SANOS–LV (reference)	3D call +1.00 NMS	13.55	13.72	15.42	62.69	8267.53
Historical SANOS (MVol, $\kappa = 1$)	1D put −1.00 NMS	5.07	5.31	7.73	73.56	4410.77
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.75 NMS	6.33	6.59	9.20	76.47	4225.99
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.50 NMS	3.12	3.38	5.97	76.05	3176.52
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.25 NMS	2.96	3.21	5.66	73.91	3014.72
Historical SANOS (MVol, $\kappa = 1$)	1D put ATM	5.59	5.94	9.48	88.85	5734.03
Historical SANOS (MVol, $\kappa = 1$)	1D call ATM	10.57	11.37	19.33	133.42	6767.86
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.25 NMS	11.65	12.41	19.97	130.20	7485.61
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.50 NMS	10.84	11.61	19.38	131.80	6854.33
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.75 NMS	12.81	13.59	21.38	132.14	8024.31
Historical SANOS (MVol, $\kappa = 1$)	1D call +1.00 NMS	8.49	9.59	20.60	156.66	6509.83
Historical SANOS (MVol, $\kappa = 1$)	3D put −1.00 NMS	4.22	4.64	8.81	96.41	3912.95
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.75 NMS	4.50	4.89	8.80	93.30	4122.77
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.50 NMS	4.63	5.01	8.81	92.03	4565.07
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.25 NMS	4.29	4.66	8.44	91.74	5014.00
Historical SANOS (MVol, $\kappa = 1$)	3D put ATM	2.66	3.03	6.75	91.00	4682.19
Historical SANOS (MVol, $\kappa = 1$)	3D call ATM	9.33	10.75	25.00	178.17	5801.27
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.25 NMS	11.12	12.66	28.02	185.09	6765.83
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.50 NMS	12.08	13.77	30.68	194.22	7356.66
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.75 NMS	11.96	13.84	32.69	204.97	7291.54
Historical SANOS (MVol, $\kappa = 1$)	3D call +1.00 NMS	11.05	13.27	35.44	222.22	7260.47

A.2 Mean–volatility

Table 9: Target-level mean–volatility scores and drawdowns.

Model/policy	Target	$\kappa = 0.1$	$\kappa = 1$	$\kappa = 2$	$\kappa = 3$	Pure RMSE	Economic MDD
SANOS–LV (reference)	1D put −1.00 NMS	13.81	87.71	169.82	251.94	82.35	4172.04
SANOS–LV (reference)	1D put −0.75 NMS	14.64	88.97	171.56	254.15	82.88	4684.10
SANOS–LV (reference)	1D put −0.50 NMS	11.10	82.38	161.58	240.78	79.28	3716.53
SANOS–LV (reference)	1D put −0.25 NMS	12.12	81.81	159.24	236.67	77.61	3536.83
SANOS–LV (reference)	1D put ATM	12.11	85.10	166.19	247.28	81.19	4226.89
SANOS–LV (reference)	1D call ATM	9.82	82.72	163.73	244.73	81.02	1877.49
SANOS–LV (reference)	1D call +0.25 NMS	13.62	92.07	179.23	266.39	87.25	3459.89
SANOS–LV (reference)	1D call +0.50 NMS	15.93	99.10	191.52	283.94	92.56	4482.35
SANOS–LV (reference)	1D call +0.75 NMS	17.45	103.83	199.80	295.78	96.20	5483.11
SANOS–LV (reference)	1D call +1.00 NMS	14.05	102.87	201.55	300.23	98.79	4163.14
SANOS–LV (reference)	3D put −1.00 NMS	16.73	58.00	103.86	149.72	47.57	7418.17
SANOS–LV (reference)	3D put −0.75 NMS	15.38	53.36	95.56	137.76	43.92	6817.57
SANOS–LV (reference)	3D put −0.50 NMS	24.67	66.49	112.96	159.43	50.80	12232.64
SANOS–LV (reference)	3D put −0.25 NMS	30.44	73.85	122.08	170.31	54.46	15649.50
SANOS–LV (reference)	3D put ATM	27.43	69.91	117.10	164.29	52.08	14016.55
SANOS–LV (reference)	3D call ATM	26.62	68.46	114.95	161.43	51.50	13426.36
SANOS–LV (reference)	3D call +0.25 NMS	26.90	69.06	115.90	162.74	51.87	13590.55
SANOS–LV (reference)	3D call +0.50 NMS	18.45	58.72	103.47	148.21	46.72	8542.56
SANOS–LV (reference)	3D call +0.75 NMS	15.71	59.71	108.60	157.49	49.78	6866.03
SANOS–LV (reference)	3D call +1.00 NMS	19.68	75.01	136.50	197.98	62.69	8267.53
Historical SANOS (MVol, $\kappa = 1$)	1D put −1.00 NMS	12.38	78.43	151.81	225.20	73.56	4410.77
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.75 NMS	13.92	82.51	158.71	234.92	76.47	4225.99
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.50 NMS	10.69	79.08	155.07	231.06	76.05	3176.52
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.25 NMS	10.32	76.79	150.64	224.49	73.91	3014.72
Historical SANOS (MVol, $\kappa = 1$)	1D put ATM	14.42	94.23	182.91	271.58	88.85	5734.03
Historical SANOS (MVol, $\kappa = 1$)	1D call ATM	23.79	143.49	276.50	409.51	133.42	6767.86
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.25 NMS	24.53	141.25	270.94	400.63	130.20	7485.61
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.50 NMS	23.89	142.11	273.47	404.84	131.80	6854.33
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.75 NMS	25.88	144.25	275.78	407.31	132.14	8024.31
Historical SANOS (MVol, $\kappa = 1$)	1D call +1.00 NMS	24.01	164.80	321.23	477.67	156.66	6509.83
Historical SANOS (MVol, $\kappa = 1$)	3D put −1.00 NMS	13.81	100.50	196.82	293.15	96.41	3912.95
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.75 NMS	13.77	97.64	190.83	284.03	93.30	4122.77
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.50 NMS	13.78	96.50	188.41	280.33	92.03	4565.07
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.25 NMS	13.41	95.89	187.53	279.17	91.74	5014.00
Historical SANOS (MVol, $\kappa = 1$)	3D put ATM	11.71	93.57	184.54	275.50	91.00	4682.19
Historical SANOS (MVol, $\kappa = 1$)	3D call ATM	26.96	187.10	365.03	542.96	178.17	5801.27
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.25 NMS	29.43	195.71	380.48	565.24	185.09	6765.83
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.50 NMS	31.28	205.75	399.60	593.46	194.22	7356.66
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.75 NMS	32.21	216.39	421.02	625.66	204.97	7291.54
Historical SANOS (MVol, $\kappa = 1$)	3D call +1.00 NMS	33.00	232.76	454.72	676.68	222.22	7260.47

A.3 Entropy OCE

Table 10: Target-level entropy oce scores and drawdowns.

Model/policy	Target	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	Pure RMSE	Economic MDD
SANOS–LV (reference)	1D put −1.00 NMS	5.63	5.93	8.86	82.35	4172.04
SANOS–LV (reference)	1D put −0.75 NMS	6.41	6.72	9.72	82.88	4684.10
SANOS–LV (reference)	1D put −0.50 NMS	3.21	3.49	6.28	79.28	3716.53
SANOS–LV (reference)	1D put −0.25 NMS	4.41	4.68	7.36	77.61	3536.83
SANOS–LV (reference)	1D put ATM	4.04	4.33	7.29	81.19	4226.89
SANOS–LV (reference)	1D call ATM	1.75	2.04	5.00	81.02	1877.49
SANOS–LV (reference)	1D call +0.25 NMS	4.95	5.29	8.68	87.25	3459.89
SANOS–LV (reference)	1D call +0.50 NMS	6.73	7.11	10.90	92.56	4482.35
SANOS–LV (reference)	1D call +0.75 NMS	7.90	8.31	12.39	96.20	5483.11
SANOS–LV (reference)	1D call +1.00 NMS	4.23	4.67	8.93	98.79	4163.14
SANOS–LV (reference)	3D put −1.00 NMS	12.15	12.25	13.25	47.57	7418.17
SANOS–LV (reference)	3D put −0.75 NMS	11.17	11.25	12.07	43.92	6817.57
SANOS–LV (reference)	3D put −0.50 NMS	20.03	20.13	21.12	50.80	12232.64
SANOS–LV (reference)	3D put −0.25 NMS	25.62	25.73	26.78	54.46	15649.50
SANOS–LV (reference)	3D put ATM	22.73	22.83	23.83	52.08	14016.55
SANOS–LV (reference)	3D call ATM	21.98	22.08	23.05	51.50	13426.36
SANOS–LV (reference)	3D call +0.25 NMS	22.23	22.33	23.32	51.87	13590.55
SANOS–LV (reference)	3D call +0.50 NMS	13.98	14.07	14.95	46.72	8542.56
SANOS–LV (reference)	3D call +0.75 NMS	10.83	10.94	12.05	49.78	6866.03
SANOS–LV (reference)	3D call +1.00 NMS	13.55	13.72	15.65	62.69	8267.53
Historical SANOS (MVol, $\kappa = 1$)	1D put −1.00 NMS	5.07	5.31	8.17	73.56	4410.77
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.75 NMS	6.33	6.59	9.57	76.47	4225.99
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.50 NMS	3.12	3.38	6.32	76.05	3176.52
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.25 NMS	2.96	3.21	5.86	73.91	3014.72
Historical SANOS (MVol, $\kappa = 1$)	1D put ATM	5.59	5.95	9.91	88.85	5734.03
Historical SANOS (MVol, $\kappa = 1$)	1D call ATM	10.57	11.42	29.60	133.42	6767.86
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.25 NMS	11.65	12.46	31.72	130.20	7485.61
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.50 NMS	10.84	11.67	32.81	131.80	6854.33
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.75 NMS	12.81	13.66	37.25	132.14	8024.31
Historical SANOS (MVol, $\kappa = 1$)	1D call +1.00 NMS	8.49	9.72	57.81	156.66	6509.83
Historical SANOS (MVol, $\kappa = 1$)	3D put −1.00 NMS	4.22	4.64	9.61	96.41	3912.95
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.75 NMS	4.50	4.89	9.30	93.30	4122.77
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.50 NMS	4.63	5.01	9.16	92.03	4565.07
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.25 NMS	4.29	4.67	8.70	91.74	5014.00
Historical SANOS (MVol, $\kappa = 1$)	3D put ATM	2.66	3.03	6.95	91.00	4682.19
Historical SANOS (MVol, $\kappa = 1$)	3D call ATM	9.33	10.91	74.55	178.17	5801.27
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.25 NMS	11.12	12.85	94.33	185.09	6765.83
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.50 NMS	12.08	14.02	123.92	194.22	7356.66
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.75 NMS	11.96	14.15	163.89	204.97	7291.54
Historical SANOS (MVol, $\kappa = 1$)	3D call +1.00 NMS	11.05	13.68	247.41	222.22	7260.47

A.4 Power-exponential OCE

Table 11: Target-level power-exponential oce scores and drawdowns.

Model/policy	Target	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	Pure RMSE	Economic MDD
SANOS–LV (reference)	1D put −1.00 NMS	5.63	5.93	8.83	82.35	4172.04
SANOS–LV (reference)	1D put −0.75 NMS	6.41	6.72	9.68	82.88	4684.10
SANOS–LV (reference)	1D put −0.50 NMS	3.21	3.49	6.24	79.28	3716.53
SANOS–LV (reference)	1D put −0.25 NMS	4.41	4.68	7.32	77.61	3536.83
SANOS–LV (reference)	1D put ATM	4.04	4.33	7.23	81.19	4226.89
SANOS–LV (reference)	1D call ATM	1.75	2.04	4.94	81.02	1877.49
SANOS–LV (reference)	1D call +0.25 NMS	4.95	5.29	8.62	87.25	3459.89
SANOS–LV (reference)	1D call +0.50 NMS	6.73	7.11	10.84	92.56	4482.35
SANOS–LV (reference)	1D call +0.75 NMS	7.90	8.31	12.32	96.20	5483.11
SANOS–LV (reference)	1D call +1.00 NMS	4.23	4.67	8.88	98.79	4163.14
SANOS–LV (reference)	3D put −1.00 NMS	12.15	12.25	13.19	47.57	7418.17
SANOS–LV (reference)	3D put −0.75 NMS	11.17	11.25	12.04	43.92	6817.57
SANOS–LV (reference)	3D put −0.50 NMS	20.03	20.13	21.09	50.80	12232.64
SANOS–LV (reference)	3D put −0.25 NMS	25.62	25.73	26.76	54.46	15649.50
SANOS–LV (reference)	3D put ATM	22.73	22.83	23.82	52.08	14016.55
SANOS–LV (reference)	3D call ATM	21.98	22.08	23.04	51.50	13426.36
SANOS–LV (reference)	3D call +0.25 NMS	22.23	22.33	23.30	51.87	13590.55
SANOS–LV (reference)	3D call +0.50 NMS	13.98	14.07	14.93	46.72	8542.56
SANOS–LV (reference)	3D call +0.75 NMS	10.83	10.94	12.00	49.78	6866.03
SANOS–LV (reference)	3D call +1.00 NMS	13.55	13.72	15.41	62.69	8267.53
Historical SANOS (MVol, $\kappa = 1$)	1D put −1.00 NMS	5.07	5.31	7.70	73.56	4410.77
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.75 NMS	6.33	6.59	9.16	76.47	4225.99
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.50 NMS	3.12	3.38	5.94	76.05	3176.52
Historical SANOS (MVol, $\kappa = 1$)	1D put −0.25 NMS	2.96	3.21	5.62	73.91	3014.72
Historical SANOS (MVol, $\kappa = 1$)	1D put ATM	5.59	5.94	9.42	88.85	5734.03
Historical SANOS (MVol, $\kappa = 1$)	1D call ATM	10.57	11.37	19.25	133.42	6767.86
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.25 NMS	11.65	12.41	19.92	130.20	7485.61
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.50 NMS	10.84	11.61	19.32	131.80	6854.33
Historical SANOS (MVol, $\kappa = 1$)	1D call +0.75 NMS	12.81	13.59	21.33	132.14	8024.31
Historical SANOS (MVol, $\kappa = 1$)	1D call +1.00 NMS	8.49	9.59	20.55	156.66	6509.83
Historical SANOS (MVol, $\kappa = 1$)	3D put −1.00 NMS	4.22	4.64	8.74	96.41	3912.95
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.75 NMS	4.50	4.89	8.72	93.30	4122.77
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.50 NMS	4.63	5.01	8.73	92.03	4565.07
Historical SANOS (MVol, $\kappa = 1$)	3D put −0.25 NMS	4.29	4.66	8.36	91.74	5014.00
Historical SANOS (MVol, $\kappa = 1$)	3D put ATM	2.66	3.03	6.66	91.00	4682.19
Historical SANOS (MVol, $\kappa = 1$)	3D call ATM	9.33	10.75	24.81	178.17	5801.27
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.25 NMS	11.12	12.66	27.87	185.09	6765.83
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.50 NMS	12.08	13.77	30.56	194.22	7356.66
Historical SANOS (MVol, $\kappa = 1$)	3D call +0.75 NMS	11.96	13.84	32.57	204.97	7291.54
Historical SANOS (MVol, $\kappa = 1$)	3D call +1.00 NMS	11.05	13.27	35.32	222.22	7260.47

References

[1] Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging. *Quantitative Finance*, 19(8): 1271–1291, 2019. doi: 10.1080/14697688.2019.1571683.

[2] Hans Buehler, Blanka Horvath, Anastasis Kratsios, Yannick Limmer, and Raeid Saqur. SANOS: Smooth arbitrage-free non-parametric option surfaces, 2026. URL https://arxiv.org/abs/2601.11209.

[3] Yilun Chen, Ami Wiesel, and Alfred O. Hero. Robust shrinkage estimation of high-dimensional covariance matrices. *IEEE Transactions on Signal Processing*, 59(9):4097–4107, 2011. doi: 10.1109/TSP.2011.2138698.

[4] Yinbin Han, Jack Yuxiang Zhang, Manuel Torres, Fernando Acero, and Renyuan Xu. Diffusion models for dynamic volatility surface generation and data-driven hedging, 2026. URL https://arxiv.org/abs/2609.13402.

[5] David E. Tyler. A distribution-free M-estimator of multivariate scatter. *The Annals of Statistics*, 15(1):234–251, 1987. doi: 10.1214/aos/1176350263.

[6] Milena Vuletić and Rama Cont. VolGAN: A generative model for arbitrage-free implied volatility surfaces. *Applied Mathematical Finance*, 31(4):203–238, 2024. doi: 10.1080/1350486X.2025.2471317.

[7] Magnus Wiese, Hans Buehler, and Blanka Horvath. GROMM: Generative risk-neutral option market models, 2026. Working paper.